

A Reputation Mechanism with Accountable Witnesses: Model, Calibration and Evaluation of the CommunityTrust Protocol (ctp Core v2.0)

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Abstract

Rotating savings groups (tontines) work without formal guarantees because their members observe each other’s past reliability. We study a digital reputation mechanism inspired by this logic: a member’s score evolves with their contributions, commitment defaults and inactivity, and a contribution only counts after approval by a randomly drawn panel of *witnesses* who stake their own score. We correct the model published in v0.1 and v1.1, which contained demonstrable errors, and derive closed-form conditions for default tolerance, panel capture by a coalition, profitability of collusion, the incentive to contribute at maximal score, the verifier’s dilemma, and the entry cost of an identity. These conditions yield a calibration method. Evaluated by simulation (up to 200 seeds per configuration) against two baselines — audit only, and witnesses without accountability —, with agents that learn to cheat and to approve without checking, and under a sensitivity analysis, the recalibrated mechanism makes collusion unprofitable for coalitions of up to half of the members, whereas audit alone lets 85 % of frauds through at the same audit rate. With adaptive agents, it keeps fraud and rubber-stamping at the exploration level, whereas the v1.1 rules, or removing witness accountability, lead to collapse (over 90 % fraud). The sensitivity analysis revealed two failure modes — audit errors and small groups —, fixed by a counter-audit and an adaptive panel, as well as an applicability condition: witness accuracy. Finally, we describe an instantiation for tontines (verifiable payments, score-based rotation order, sponsorship) and a data collection protocol for its empirical calibration, which remains to be carried out.

Keywords: reputation systems, tontines, ROSCA, collusion, Sybil attack, verifier’s dilemma, mechanism design, agent-based simulation.

1 Introduction

Rotating savings and credit associations (ROSCAs), called tontines in West Africa, gather members who periodically pay a contribution whose total is allocated in turn to one of them [3, 5, 13]. Their structural weakness is well known: a member who has already received the pot has an incentive to stop contributing, and enforcement relies on social pressure and the threat of exclusion [2]. Reputation — the collective memory of past behaviour — is therefore the central mechanism.

Digital reputation systems try to reproduce this memory at scale [22, 8, 15]. They face well-documented difficulties: manipulation by coalitions, multiple identities (Sybil attack [9]),

reputation laundering through identity change [10], and verifier laziness when checking is costly [19].

The CommunityTrust Protocol (CTP) proposes a mechanism in which a contribution only counts after approval by *witnesses* who risk their own score. Versions v0.1 and v1.1 claimed universality and strategic robustness; a review showed that these claims were not established (Appendix A).

Contributions.

1. A consistent formal model with explicit assumptions (Section 3).
2. Nine propositions and two corollaries, including seven closed-form conditions usable for calibration (Section 4).
3. A parameter set derived from these conditions (Section 5).
4. A reproducible simulation study: verification of the conditions, comparison with two baselines, adaptive agents and sensitivity analysis (Section 6).
5. An instantiation for tontines with two results on rotation order and sponsorship (Section 7).
6. An erratum of the previous versions and a data collection protocol (Appendices A and B).

What this paper does not claim. The idea of a continuous, dynamic, non-transferable trust score is not new (Section 2). The results concern a model; no field data is used yet. The Node hierarchy, global score and portability of previous versions are out of scope (Appendix C).

2 Related work

Computational trust and reputation. Marsh [20] formalises trust as a continuous quantity updated by experience. Resnick et al. [22], Dellarocas [8] and Jøsang et al. [15] describe rating-based mechanisms and their failure modes. EigenTrust [16] aggregates local ratings into a global score; PeerTrust [25] weights feedback by the credibility of its source — a principle CTP reuses by weighting a contribution’s quality by the score of its approvers.

Attacks. Without an authority certifying identities, an attacker can always present multiple identities [9], and no symmetric reputation function is robust to this [7]. When pseudonyms are free, newcomers must start at the minimal score, otherwise identity change erases sanctions [10]. Hoffman et al. [14] survey these attacks and their defences.

Incentivised verification. Luu et al. [19] describe the *verifier’s dilemma*: when checking is costly and fraud is rare, a rational verifier accepts without checking. TrueBit [23] addresses it by injecting deliberate errors (*forced errors*) that verifiers are rewarded for detecting. Kleros [18] draws jurors at random who post a deposit and are rewarded when voting with the majority. CTP combines these ideas: random panel, majority reward, score-proportional sanction, decoy signals.

Repeated games and community norms. Sustaining cooperation through the threat of future sanctions falls under the folk theorem [11]; Kandori [17] studies sanctions exercised by the community. The “dominance” conditions of v1.1 were instances of these results and are not claimed as contributions.

Tontines and joint liability. Besley, Coate and Loury [5] and Handa and Kirton [13] analyse the economics of ROSCAs; Anderson, Baland and Moene [2] study enforcement after the pot has been received. Abebe et al. [1] study ROSCAs from an algorithmic perspective. Joint liability in microcredit [4, 12] underpins the sponsorship mechanism of Section 7. Ostrom [21] studies the governance of commons without an external authority.

Non-transferable reputation. Critiques of token-weighted voting [6] and non-transferable tokens [24] share the goal of decoupling influence from wealth.

3 Model

3.1 Agents, periods and signals

A Node (community) has n members. Time is discrete, $t \in \mathbb{N}$. Each member i has a score $\tau_i(t) \in [0, 1]$. In each period, a member either *defaults* on their commitment, submits a signal s (candidate contribution), or does nothing. Each signal has a true conformity $Q(s) \in \{0, 1\}$ that the protocol does not observe directly.

3.2 Witness panel and acceptance

A member is eligible as a witness if $\tau_w(t) \geq \tau_W$. For each signal, a panel $\mathcal{P}(s)$ of k witnesses is drawn uniformly without replacement among eligible members other than the author, with $k = \max(k_{\min}, \lfloor \alpha_k(n - 1) \rfloor)$. Each witness approves or rejects; $\mathcal{A}(s)$ denotes the approvers. The signal is accepted if $|\mathcal{A}(s)| \geq a_k = \lceil \theta k \rceil$. If fewer than k members are eligible, the panel is reduced to all m eligible members when $m \geq 3$ (with $a_m = \lceil \theta m \rceil$); if $m < 3$, the signal is provisionally accepted and audited with certainty. Without this rule, a small Node in which a few members fall below τ_W can no longer evaluate any signal (Section 6, E7). The measured quality of an accepted signal is

$$q(s) = \bar{\tau}_{\mathcal{A}(s)} \cdot \frac{|\mathcal{A}(s)|}{k}. \quad (1)$$

3.3 Audit and decoy signals

Each accepted contribution is independently audited with probability p_a ; the audit reveals $Q(s)$ with error probability ε_a . A non-conformity verdict triggers an independent *counter-audit*; the contribution is *sanctioned* only if both verdicts agree. The probability of sanctioning a conforming contribution thus drops from ε_a to ε_a^2 , at the cost of letting an audited non-conforming contribution through with probability at most $2\varepsilon_a$. In addition, in each period and for each member, the protocol injects a *decoy signal* with probability h : a signal known to be non-conforming, presented to a panel like an ordinary one.

3.4 Score dynamics

$$\tau_i(t + 1) = \text{clip}(\tau_i + G_i + R_i - L_i - S_i - D_i - W_i, 0, 1) \quad (2)$$

$$G_i = \eta \sum_{s: \text{author}=i, \text{ accepted, not sanctioned}} q(s) \quad \text{author gain} \quad (3)$$

$$R_i = \frac{\eta_W}{k} \cdot \#\{s : i \in \mathcal{P}(s), \text{ vote of } i = \text{outcome of } s\} \quad \text{majority reward} \quad (4)$$

$$L_i = \kappa \mathbf{1}\{i \text{ defaults}\} \quad \text{commitment default} \quad (5)$$

$$S_i = \kappa_s \cdot \#\{s : \text{author} = i, \text{ sanctioned}\} \quad \text{author sanction} \quad (6)$$

$$D_i = \delta \tau_i \mathbf{1}\{i \text{ inactive}\} \quad \text{depreciation} \quad (7)$$

$$W_i = \phi \tau_i \cdot \#\{s : i \in \mathcal{A}(s), s \text{ sanctioned or decoy}\} \quad \text{witness accountability} \quad (8)$$

The *outcome* of a signal is either “accepted and not sanctioned” or “rejected or sanctioned”; a decoy always has the latter outcome. A member is *inactive* if they have neither an accepted, non-sanctioned contribution nor a default in the period: submitting a rejected signal is not enough to remain active. The reward R_i is only paid to a member who is active in the period: it cannot compensate for not contributing.

Differences from v1.1 and rationale. (i) R_i is new: without it, validating brings nothing and carries a risk. It rewards agreement with the outcome rather than approval, so as not to bias votes towards approval. (ii) S_i is new: without it, submitting a fraud is riskless. (iii) The witness sanction is no longer divided by $|\mathcal{A}(s)|$: the division diluted accountability as accomplices became more numerous. (iv) Auditing is uniform instead of being restricted to signals with low $q(s)$. (v) Inactivity is defined by the absence of an accepted contribution. (vi) Decoy signals are new (Prop. 8). (vii) The witness reward is conditional on activity: without this rule, a member who never contributes could keep their score by serving only as a witness (observed in simulation: stationary score of about 0.3 for free riders). (viii) The counter-audit and panel reduction are new; both fix failures observed in simulation (E7).

3.5 Assumptions

Assumption 1 (Audit). An audit reveals the true conformity, with error probability ε_a (zero unless stated otherwise).

Assumption 2 (Non-manipulable draw). The panel is drawn uniformly; neither the author nor the witnesses choose it, and decoy signals are indistinguishable from ordinary ones.

Assumption 3 (Entry). Founding members start at τ_0 ; every newcomer starts at 0, unless sponsored (Section 7).

Assumption 4 (Utility). For the strategic analysis, utility is linear in score, away from the bounds, minus effort costs: c_h for a conforming contribution, c_v for a check.

4 Analysis

Proposition 1 (Boundedness). *If $\tau_i(0) \in [0, 1]$ for all i , then $\tau_i(t) \in [0, 1]$ for all t .*

Proof. By induction, since clip takes values in $[0, 1]$. □

Proposition 2 (Saturation). *If a member obtains in every period an accepted, non-sanctioned contribution of quality $q(s) \geq \underline{q} > 0$, with no negative term, then $\tau_i(t) = 1$ for all $t \geq \lceil (1 - \tau_i(0))/(\eta \underline{q}) \rceil$.*

Proof. Each period adds at least $\eta \underline{q}$ as long as $\tau_i < 1$. □

Proposition 3 (Decay under inactivity). *A member inactive over $[0, t]$ satisfies $\tau_i(t) = \tau_i(0)(1 - \delta)^t$.*

4.1 Default tolerance

Proposition 4 (Tolerance threshold). *Consider a member who defaults with probability p_d per period and otherwise obtains an accepted, conforming contribution of mean quality $\bar{q}$. For $\tau_i \in [\kappa, 1 - \eta]$, $\mathbb{E}[\Delta \tau_i] = (1 - p_d)\eta \bar{q} - p_d \kappa$, which is strictly positive if and only if*

$$p_d < p_d^* = \frac{\eta \bar{q}}{\eta \bar{q} + \kappa}. \quad (9)$$

Proof. In this interval no bound is reached within one period; default and contribution are mutually exclusive; take expectations. $\square$

Corollary 1 (Calibrating κ). *For a target tolerance p^* , one needs $\kappa = \eta\bar{q}(1 - p^*)/p^*$: the ratio κ/η , not the absolute values, sets the tolerance.*

4.2 Panel capture

Proposition 5 (Capture probability). *Let there be m eligible members other than the author, including c accomplices who unconditionally approve the coalition's signals, the others rejecting non-conforming signals. The probability that a non-conforming coalition signal is accepted is at least*

$$\pi(m, c, k, a_k) = \sum_{x=a_k}^{\min(c, k)} \frac{\binom{c}{x} \binom{m-c}{k-x}}{\binom{m}{k}}, \quad (10)$$

with equality if honest witnesses never err.

Proof. Under Assumption 2, the number of accomplices in the panel follows a hypergeometric distribution (m, c, k) ; honest witnesses' errors can only add approvals. $\square$

4.3 Profitability of collusion

Proposition 6 (Audit threshold). *Consider a non-conforming coalition contribution, accepted with quality q by $x \geq a_k$ accomplices of mean score $\bar{\tau}$. The coalition's expected gain is $\mathbb{E}[\Delta] = (1 - p_a)(\eta q + x\eta_W/k) - p_a(\kappa_s + \phi x\bar{\tau})$, which is negative if and only if*

$$p_a > p_a^* = \frac{\eta q + x\eta_W/k}{\eta q + x\eta_W/k + \kappa_s + \phi x\bar{\tau}}. \quad (11)$$

Since $q \leq 1$, $x\eta_W/k \leq \eta_W$, $x \geq a_k$ and $\bar{\tau} \geq \tau_W$, $p_a^ \leq (\eta + \eta_W)/(\eta + \eta_W + \kappa_s + \phi a_k \tau_W)$.*

Proof. If not sanctioned, the contribution yields ηq to the author and η_W/k to each approving accomplice; if sanctioned, it costs κ_s to the author and $\phi\tau_w$ to each approver. The bound follows from the monotonicity of p_a^* in each term. $\square$

Corollary 2 (Collusion without audit). *Assume $p_a = 0$ and a homogeneous coalition with score τ , each of whose non-conforming signals is accepted with probability π and rejected otherwise. If inactivity is defined by the absence of an accepted contribution, the expected per-period score change of a colluder is at most $\pi\eta\tau - (1 - \pi)\delta\tau$ (away from the bounds, excluding witness rewards), which is negative as soon as*

$$\frac{\pi}{1 - \pi} < \frac{\delta}{\eta}. \quad (12)$$

With $\delta = \eta$, collusion only pays if the coalition captures more than half of the panels. If inactivity is defined by the absence of submission (v1.1 rule), a rejected signal costs nothing and collusion pays for any $\pi > 0$.

Proof. If accepted (probability π), the signal yields $\eta q \leq \eta\tau$ since $q = \bar{\tau}_A|\mathcal{A}|/k \leq \tau$ for approvers with score τ ; if rejected, the colluder is inactive and loses $\delta\tau$. $\square$

Proposition 6 remains necessary when π exceeds this threshold — a majority coalition — or when depreciation is small relative to the gain.

4.4 Incentive at maximal score

Proposition 7 (Ceiling). *A member at score $\tau_i = 1$ gains nothing from an accepted contribution. Under Assumption 4, contributing honestly is better than submitting a non-conforming signal (rejected with probability $1 - \pi_1$) if*

$$c_h < (1 - \pi_1) \delta + \pi_1 p_a \kappa_s, \quad (13)$$

where π_1 is the probability that an isolated non-conforming signal is accepted. If inactivity is defined by the absence of submission (v1.1 rule), a saturated member is never inactive, the right-hand side reduces to $\pi_1 p_a \kappa_s$, and honest contribution is no longer incentivised once $c_h > \pi_1 p_a \kappa_s$.

Proof. At $\tau_i = 1$, G_i is absorbed by the ceiling. An honest contribution costs c_h and avoids depreciation δ . A non-conforming signal costs nothing; if rejected (probability $1 - \pi_1$) it leaves the member inactive (loss δ); if accepted it avoids depreciation but exposes the member to the sanction κ_s with probability p_a . $\square$

4.5 Verifier's dilemma

Proposition 8 (Decoy signals). *A witness with score τ_w can check (cost c_v) or rubber-stamp (approve without checking). Let h' be the share of decoy signals among the signals presented. Counting decoys only, rubber-stamping is strictly dominated if*

$$h' (\phi \tau_w + \eta_W / k) > c_v. \quad (14)$$

Without decoys ($h' = 0$) and if real frauds are rare, rubber-stamping is the best response: this is the verifier's dilemma.

Proof. On a conforming signal, checking and rubber-stamping lead to the same vote (approval) and the same reward; only c_v differs. On a decoy, rubber-stamping costs $\phi \tau_w$ and forfeits η_W / k , whereas checking leads to rejection (barring error). Real non-conforming signals only add to the advantage of checking. $\square$

4.6 Entry cost

Proposition 9 (Eligibility delay). *Under Assumption 3, a new unsponsored identity cannot be a witness before $\lceil \tau_W / \eta \rceil$ periods, each including an accepted contribution.*

Proof. A non-witness member gains at most $\eta q \leq \eta$ per period, and nothing without an accepted contribution. $\square$

The mechanism therefore does not make Sybil attacks impossible — which is ruled out for a symmetric function [7] — but gives them a cost linear in the number of identities, which is deterrent only if an accepted contribution is itself costly. Since the score is bounded below by 0, the effective sanction of a default is $\min(\kappa, \tau_i)$; Section 7 addresses this limit through exclusion and sponsorship.

5 Calibration

Table 1 derives the recommended parameters (v2.1) from the propositions above. The v1.1 parameters, kept for comparison, are labelled v2.0 when used with the corrected model.

Two parameter-free rules are added: the counter-audit and the adaptive panel (Section 3). The numerical rationales use $\bar{q} = 0.91$ (measured in simulation) and, for Proposition 8, $c_v = 0.001$ and $\tau_w = \tau_W$. For the v1.1 parameters, the same Proposition 4 gives $p_d^* = 0.23$: a member can default about one period in four while seeing their score increase.

Table 1: Parameters: original v1.1 and recommended v2.1.

	Role	v1.1	v2.1	Rationale for v2.1
η	author gain	0.05	0.02	saturation in ≈ 55 periods (Prop. 2)
κ	default penalty	0.15	0.35	$p_d^* \approx 0.05$ (Cor. 1)
η_W	witness reward	—	0.005	small relative to η
κ_s	author sanction	—	0.10	
ϕ	witness accountability	0.15	0.30	$p_a^* \leq 0.052$ (Prop. 6)
p_a	audit probability	0.15	0.15	$\approx 3\times$ margin over p_a^*
$k_{\min}$	minimal panel size	3	5	$\pi \approx 0.006$ for 30 % accomplices, $n = 20$
θ	approval fraction	1	2/3	tolerance to witness errors
h	decoy rate	—	0.05	$h' \approx 0.048 > 0.011$ (Prop. 8)
δ	depreciation	0.02	0.02	$> c_h$ required (Prop. 7)
α_k	panel fraction	0.20	0.20	
τ_W	eligibility threshold	0.30	0.30	
τ_0	founders' score	—	0.50	choice (sensitivity in E7)

6 Simulation study

6.1 Protocol

The simulations implement exactly the model of Section 3 over $T = 104$ periods (300 for E6). 95% confidence intervals are obtained by bootstrap over seeds. Behaviours: *honest* (true quality $\sim \mathcal{N}(0.72; 0.12)$); *free rider* (active with probability 0.08, quality $\sim \mathcal{U}(0.10; 0.35)$); *isolated fraudster* (quality $\sim \mathcal{N}(0.20; 0.08)$, honest witness); *colluder* (like the fraudster, but unconditionally approves other colluders' signals); *adaptive* (E6). A signal is conforming if its true quality is at least 0.45; a checking witness observes it with noise $\mathcal{N}(0; \sigma_{\text{obs}})$, $\sigma_{\text{obs}} = 0.10$. Three rule sets are compared: **v1.1** (original rules and parameters), **v2.0** (corrected model, v1.1 parameters, no decoys, divided sanction, inactivity defined by absence of submission), **v2.1** (corrected model, Table 1 parameters).

6.2 E1 — Default tolerance

Twenty honest members, five of whom default with probability p_d (Figure 1).

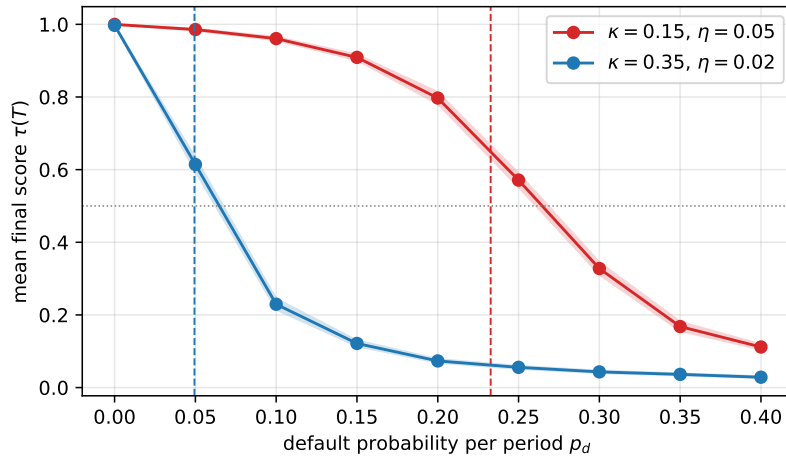


Figure 1: E1 — Final score of defaulting members (95 % CI, 200 seeds). Dashed: threshold p_d^* (Prop. 4); dotted: initial score.

With the v1.1 parameters ($p_d^* = 0.23$), members defaulting one period in seven end at

0.91 [0.90; 0.92], and the final score only drops below the initial score between $p_d = 0.25$ and $p_d = 0.30$. With the v2.1 parameters ($p_d^* = 0.049$), defaulting one period in twenty leaves the score at 0.61 [0.60; 0.63], slightly above the initial score as expected near the threshold, and defaulting one period in ten brings it down to 0.23 [0.22; 0.25]. Both observed thresholds agree with (9) up to the offset due to the variation of $\bar{q}$ and to the bounds.

6.3 E2 — Collusion and audit probability

Coalitions of 8 and 12 members out of 20 (40 % and 60 %) (Figure 2).

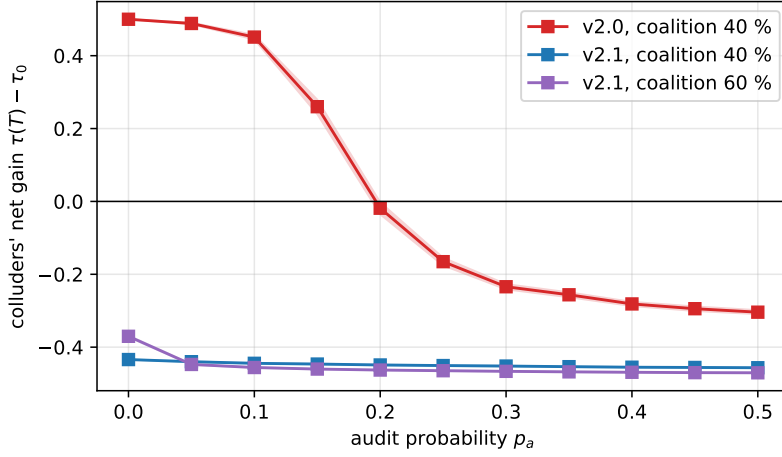


Figure 2: E2 — Mean net gain of a coalition as a function of p_a (95 % CI, 200 seeds).

With the v1.1 parameters (40 % coalition), the net gain is +0.50 without audit, +0.26 [0.24; 0.28] at $p_a = 0.15$, no longer significantly non-zero at $p_a = 0.20$ (-0.02 [-0.04 ; 0.00]), and negative beyond, in line with the threshold of Proposition 6 in its divided-sanction form (0.146 to 0.194). With v2.1, collusion loses *for every value of p_a , including zero*: -0.43 without audit for a 40 % coalition and -0.37 for a 60 % coalition. This is what Corollary 2 predicts: with $\delta/\eta = 1$, collusion requires $\pi > 1/2$, whereas the capture probability is about 0.04 at 40 % (E3) and $\pi(19, 11, 5, 4) \approx 0.27$ at 60 %. In v2.1, it is the inactivity rule, not the audit, that deters minority collusion; the audit remains the protection against a majority coalition and against careless witnesses (E6).

6.4 E3 — Panel capture

Acceptance rate of a coalition’s non-conforming signals, $n = 20$, no audit (Figure 3).

With the v1.1 parameters ($k = 3$, $a_k = 2$), a 30 % coalition gets 17 % of its non-conforming signals accepted, and a half-size coalition 49 %. With v2.1 ($k = 5$, $a_k = 4$), these rates drop to 1.8 % and 12.7 %. Measured rates exceed the bound (10) by the effect of honest witnesses’ errors, up to sampling noise (45 %).

6.5 E4 — Comparison of the three rule sets

Twenty members: 10 honest ($p_d = 0.02$), 4 free riders, 6 colluders (Table 2, Figure 4).

With v1.1, colluders end above their initial score (0.58): their rare captured contributions have $q(s)$ above $q_{\min}$ and are never audited, and unanimity wrongly rejects 8.6 % of honest contributions. The corrected model with v1.1 parameters (v2.0) is *worse*: 19 % of frauds are accepted and colluders reach 0.78, because 2-out-of-3 approval multiplies the capture probability by fifteen ($\pi \approx 0.010 \rightarrow 0.155$), p_a stays below threshold, and a rejected signal

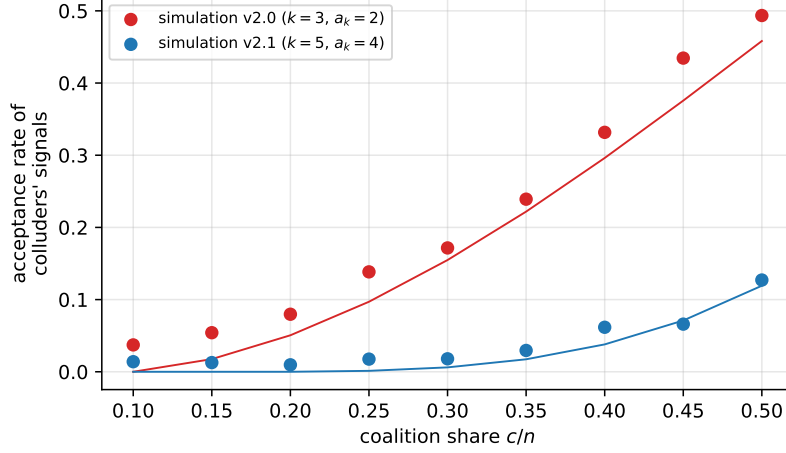


Figure 3: E3 — Dots: simulation (100 seeds); lines: hypergeometric bound of Proposition 5.

Table 2: E4 — Mean final score per group and error rates (200 seeds; 95 % CI in brackets).

	v1.1	v2.0	v2.1
Honest	0.99 [0.99; 0.99]	0.99 [0.99; 0.99]	0.91 [0.91; 0.92]
Free riders	0.07 [0.07; 0.07]	0.07 [0.07; 0.07]	0.06 [0.06; 0.06]
Colluders	0.58 [0.57; 0.58]	0.78 [0.77; 0.80]	0.06 [0.06; 0.06]
Frauds accepted	2.5 %	19.1 %	0.5 %
Wrongful rejections	8.6 %	1.6 %	3.7 %
Votes per signal	3.0	3.0	5.3
Audits per signal	0.00	0.10	0.09

costs nothing. With v2.1, colluders fall to the free riders' level (0.06) and 0.5 % of frauds pass. The trade-off is a mean honest score of 0.91 instead of 0.99: a harsher default penalty and 3.7 % wrongful rejections make accidental defaults ($p_d = 0.02$) costly. This cost follows directly from the choice of p_d^* and can be tuned via Corollary 1.

6.6 E5 — Do witnesses add anything?

To isolate the contribution of witnesses, v2.1 is compared with two baselines at the same $p_a = 0.15$: *audit only* (every contribution is accepted with $q = 1$, and only the audit sanctions) and *witnesses without accountability* ($\phi = \kappa_s = \eta_W = h = 0$). Population: 13 honest members, 3 isolated fraudsters, 4 colluders (Table 3).

Table 3: E5 — Comparison with baselines (200 seeds).

	audit only	no accountability	v2.1
Honest	1.00 [1.00; 1.00]	1.00 [1.00; 1.00]	1.00 [1.00; 1.00]
Isolated fraudsters	0.52 [0.50; 0.54]	0.06 [0.06; 0.06]	0.06 [0.06; 0.06]
Colluders	0.53 [0.51; 0.55]	0.06 [0.06; 0.06]	0.06 [0.06; 0.06]
Frauds accepted	84.9 %	0.2 %	0.2 %
Wrongful rejections	0.0 %	3.8 %	3.7 %
Votes per signal	0.0	5.0	5.2
Audits per signal	0.15	0.09	0.09

Audit alone, at $p_a = 0.15$ (Figure 5), lets 85 % of frauds through: fraudsters keep their initial score. Bringing this rate down to 5 % would require auditing 95 % of contributions,

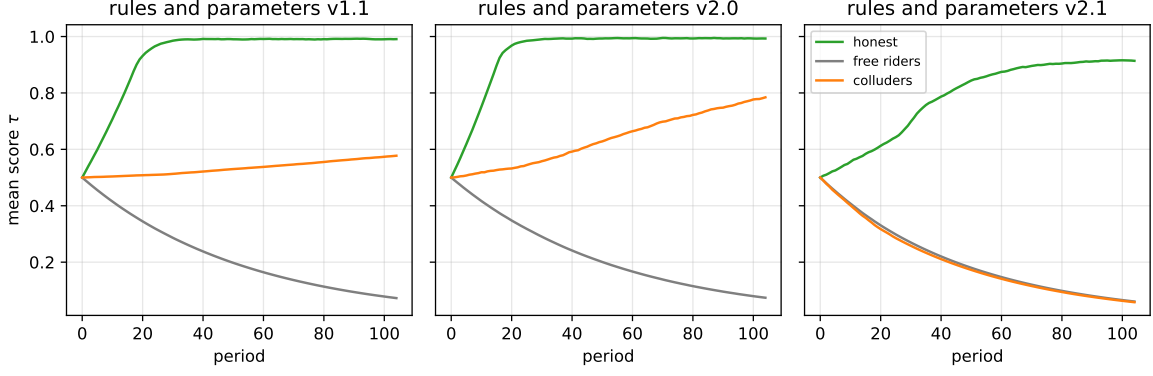


Figure 4: E4 — Mean score trajectories per group.

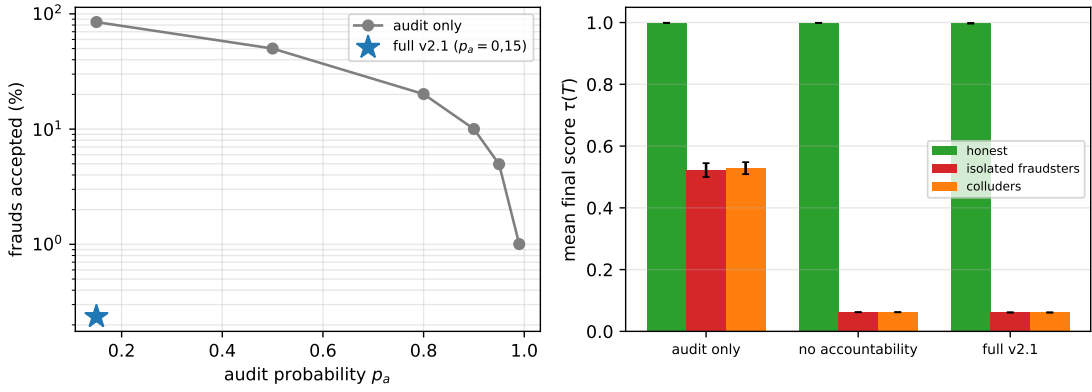


Figure 5: E5 — Left: share of frauds accepted by audit alone as a function of p_a (log scale), compared with v2.1 at $p_a = 0.15$. Right: final score per group (95 % CI).

and 99 % to reach 1 %. The witness panel achieves 0.24 % with 5 votes and 0.09 audits per signal: its value depends on the relative cost of a vote and an audit, and grows with the cost of auditing. On the other hand, with fixed behaviours, witnesses *without* accountability do exactly as well as v2.1. This is not surprising: accountability acts on incentives, which fixed-behaviour agents do not have. E6 tests this point.

6.7 E6 — Adaptive agents

The fixed behaviours of E1–E5 do not test incentives. In E6, all 20 members are adaptive: in each period, each one chooses as author between contributing honestly (cost $c_h = 0.01$) and cheating, and as witness between checking (cost $c_v = 0.001$) and rubber-stamping. Choices are learned with ε -greedy ($\varepsilon = 0.1$, step 0.1) on the realised payoff of each role, over 300 periods and 50 seeds. Besides the three rule sets, two ablations of v2.1 are tested: without decoys ($h = 0$), and without accountability ($\phi = \kappa_s = \eta_W = h = 0$) (Figure 6).

With the v1.1 rules, the system collapses: 95 % of authors cheat, 95 % of witnesses rubber-stamp, and 87 % of frauds are accepted. With v2.0, witnesses mostly check (14 % rubber-stamping), but 68 % of authors submit non-conforming signals: saturated at $\tau = 1$, they gain nothing by contributing honestly, and a rejected signal does not make them inactive (Prop. 7). With v2.1, fraud and rubber-stamping stay at the exploration level (7.1 % [6.3; 8.2] and 5.5 % [5.0; 6.0]) and 2.1 % of frauds pass. The two ablations confirm the role of each component: without decoys, 31 % of witnesses rubber-stamp and 7.5 % of frauds pass (Prop. 8); without witness accountability and author sanction, the system collapses as in v1.1 (92 % fraud,

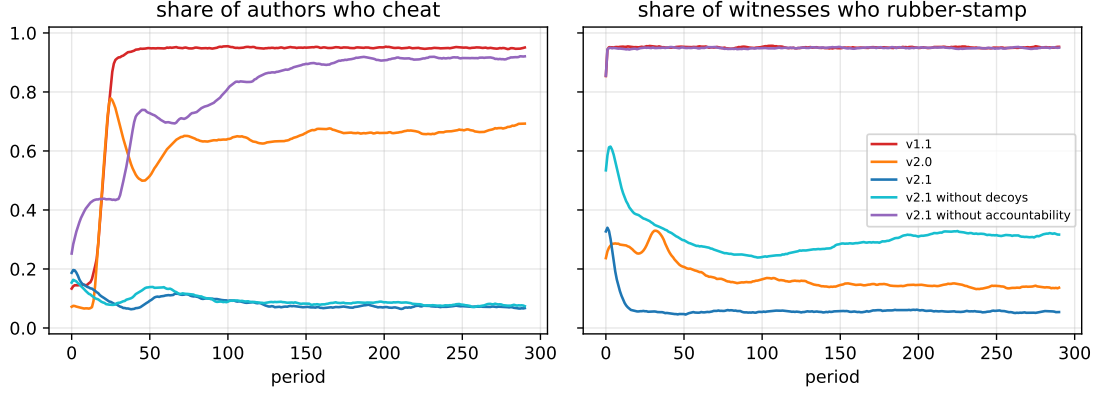


Figure 6: E6 — Share of authors who cheat and of witnesses who rubber-stamp (mean over 50 seeds, 10-period moving average). The floor of about 0.05 corresponds to $\varepsilon/2$ exploration.

95 % rubber-stamping, 83 % of frauds accepted). Accountability, seemingly useless with fixed behaviours (E5), is what prevents collapse when agents adapt.

6.8 E7 — Sensitivity

One factor at a time is varied around the v2.1 configuration (20 members including 30 % colluders and 10 % free riders, 100 seeds). Table 4 reports the honest members’ final score, the colluders’ net gain, and error rates.

Table 4: E7 — Sensitivity of v2.1 (100 seeds per row).

Variant	honest	colluders’ gain	frauds accepted	wrongful rejections
v2.1 reference	0.92 [0.91; 0.93]	−0.44	0.4 %	3.7 %
$\sigma_{\text{obs}} = 0.05$	0.93 [0.92; 0.94]	−0.44	0.3 %	1.2 %
$\sigma_{\text{obs}} = 0.20$	0.70 [0.68; 0.72]	−0.45	1.3 %	14.1 %
$\sigma_{\text{obs}} = 0.20, \theta = 0.5$	0.73 [0.71; 0.75]	−0.46	5.2 %	3.9 %
$\varepsilon_a = 0.05$	0.91 [0.90; 0.92]	−0.44	0.4 %	3.8 %
$\varepsilon_a = 0.05$, no counter-audit	0.62 [0.59; 0.65]	−0.45	0.4 %	3.8 %
$\varepsilon_a = 0.10$	0.87 [0.86; 0.89]	−0.44	0.5 %	3.8 %
$\varepsilon_a = 0.10$, no counter-audit	0.36 [0.33; 0.39]	−0.48	1.1 %	3.2 %
$\tau_0 = 0.3$	0.92 [0.91; 0.93]	−0.26	0.2 %	3.8 %
$\tau_0 = 0.7$	0.92 [0.91; 0.92]	−0.62	0.6 %	3.8 %
$n = 10$	0.92 [0.90; 0.93]	−0.44	0.3 %	3.6 %
$n = 40$	0.91 [0.91; 0.92]	−0.44	0.6 %	2.4 %
$n = 10$, non-adaptive panel	0.36 [0.28; 0.45]	−0.44	0.2 %	2.1 %
coalition 10 %	0.90 [0.89; 0.91]	−0.44	0.6 %	3.7 %
coalition 50 %	0.93 [0.92; 0.94]	−0.46	1.3 %	3.7 %
$\theta = 0.5$	0.92 [0.91; 0.93]	−0.45	2.3 %	1.1 %
$\theta = 1$	0.88 [0.87; 0.89]	−0.44	0.0 %	12.0 %
$p_a = 0.05$	0.92 [0.91; 0.93]	−0.44	0.6 %	3.8 %
$p_a = 0.30$	0.92 [0.91; 0.93]	−0.44	0.3 %	3.7 %
$h = 0$	0.92 [0.91; 0.93]	−0.44	0.4 %	3.8 %
$h = 0.10$	0.91 [0.90; 0.92]	−0.44	0.5 %	3.8 %

The mechanism (Figure 7) is stable with respect to the initial score, Node size, coalition size up to 50 % (1.3 % of frauds accepted), p_a between 0.05 and 0.30, and the decoy rate. Two failure modes, revealed by a first version of this analysis, led to changes in the model: without counter-audit, a 5 % audit error brings honest members down to 0.62 and a 10 % error to 0.36, because every wrongful sanction removes $\phi\tau_w$ from all approvers and pushes honest members

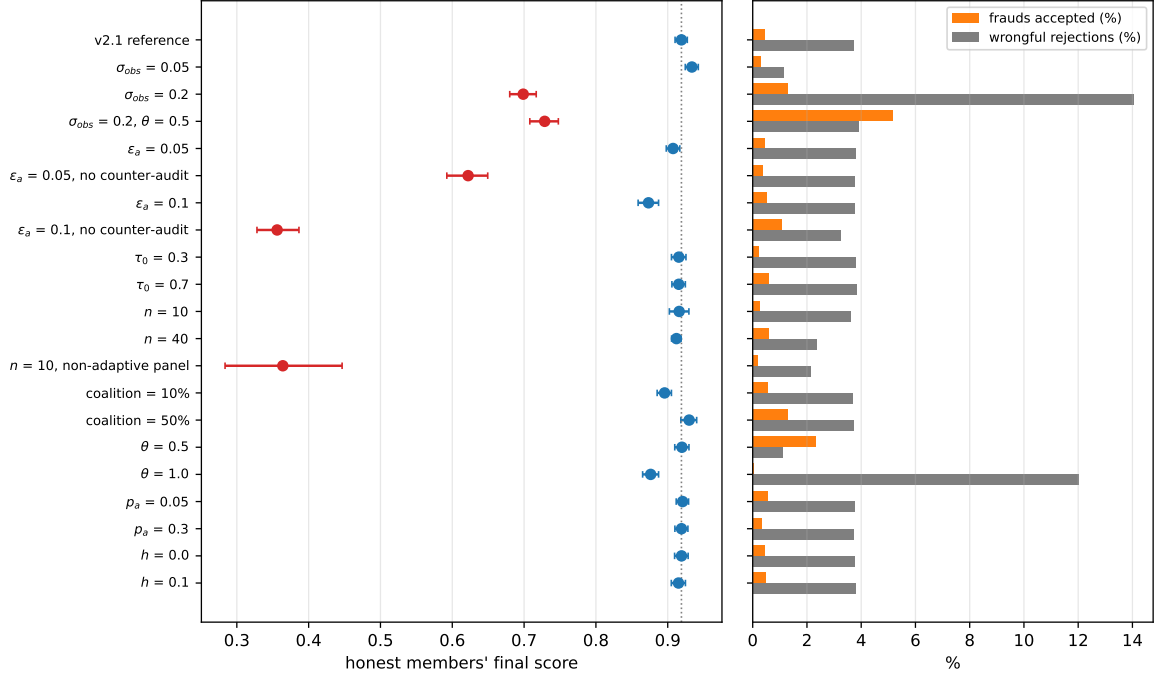


Figure 7: E7 — Left: honest members' final score per variant (95 % CI; red below 0.8). Right: frauds accepted and wrongful rejections.

below τ_W ; with the counter-audit, they stay at 0.91 and 0.87. In a 10-member Node, a fixed-size panel brings honest members down to 0.36 for lack of eligible witnesses; the adaptive panel keeps them at 0.92.

The remaining sensitivity concerns witness accuracy. With $\sigma_{obs} = 0.20$, 14% of honest contributions are wrongly rejected and the honest score drops to 0.70. Lowering θ to 0.5 brings wrongful rejections down to 3.9% but lets 5.2% of frauds through, and the honest score only recovers to 0.73: inaccurate witnesses themselves approve non-conforming signals and are sanctioned. No tuning compensates for witnesses unable to judge; this is an applicability condition of the mechanism (Section 8).

7 Instantiation for tontines

In a tontine, the main signal — a contribution payment — is verifiable from the payment operator's records. Auditing is therefore exhaustive and nearly free ($p_a = 1$, $\epsilon_a \approx 0$), and the witness panel is unnecessary: set $q = 1$ for every payment made on time. Witnesses remain useful for commitments that cannot be verified automatically (e.g. an in-kind contribution). The risk specific to tontines, a member leaving after receiving the pot [2], is handled by two rules.

Rule 1 (Score-based rotation). Pot positions are assigned by decreasing score; ties are broken by seniority.

Proposition 10 (Exposure). *In a tontine of N members with contribution m , the member receiving the pot at position r can, by stopping payments, appropriate at most $(N - r)m$. Under the score-based rotation rule, a member whose score is strictly lower than all others receives the pot last and has zero exposure.*

Proof. At position r , the member has paid rm and received Nm ; $(N - r)m$ remains to be paid. At position N , nothing remains. $\square$

Rule 2 (Sponsorship and exclusion). A newcomer may only join with a sponsor whose score is at least τ_W . They start at τ_p . During a probation of P periods, each default of the sponsored member costs ψ to the sponsor. A member whose score reaches 0 after a default is excluded; readmission requires a new sponsor.

Proposition 11 (Defaults before exclusion). *Under this rule, a sponsored identity with C accepted contributions in total commits at most $1 + \lfloor (\tau_p + \eta C)/\kappa \rfloor$ defaults before exclusion (excluding witness rewards), and each default during probation costs ψ to its sponsor. With $\kappa = 0.35$ and $\tau_p \leq 0.35$, an identity that defaults before any contribution is excluded at its first default.*

Proof. The cumulative available score is at most $\tau_p + \eta C$; each default removes κ while the score remains positive, and the default that brings it to 0 triggers exclusion. $\square$

Sponsorship shifts the cost of a new identity — and hence of Sybil attacks and of laundering through identity change — onto an established member, who is thus incentivised to screen candidates. This is the logic of joint liability in microcredit [4, 12], and that of guarantors in traditional tontines. Combined with score-based rotation, it places newcomers at the end of the rotation, where their exposure is zero, until they have built a track record.

8 Discussion

What the results establish. The revised mechanism admits closed-form conditions confirmed by simulation, and these conditions make it possible to calibrate it. Each component has an identifiable role: the panel filters frauds (E3, E5); the inactivity rule makes minority collusion unprofitable even without audit (Cor. 2, E2) and preserves the incentive at maximal score (Prop. 7, E6); witness accountability and decoys prevent rubber-stamping (Prop. 8, E6); the audit, backed by a counter-audit, protects against majority coalitions without penalising honest members (E7).

What witnesses cost. The panel replaces near-exhaustive auditing with five votes per signal. It is only worthwhile if a vote costs much less than an audit.

Applicability condition. The mechanism assumes witnesses able to judge conformity with good accuracy (E7). When this is not the case, it is better to score only automatically verifiable contributions — which is precisely the case of tontines (Section 7), where witnesses become unnecessary.

Scope. Previous versions presented CTP as universal. The results rather show a parameterised framework whose relevant parameters (κ/η , δ/η , k , θ , h) depend on the nature of contributions, on witness accuracy and on the cost of an identity; each instantiation must be calibrated, then evaluated on real data.

9 Limitations

- **No real data.** Quality and default distributions and the costs c_h , c_v are assumed. Appendix B describes the required data collection protocol.
- **Simple learning.** The E6 agents learn independently; a coalition learning jointly, or an adversary trying to recognise decoys, is not modelled. Decoy indistinguishability (Assumption 2) is a design requirement, not a given.

- **Witness accuracy.** The mechanism degrades when witnesses judge poorly (E7); measuring this accuracy is a prerequisite for any application to non-verifiable contributions.
- **Audit organisation.** The model fixes p_a and ε_a without specifying who audits; outside the case of payments, this remains an open question.
- **Utility.** The strategic analysis assumes utility linear in score; the actual form of the rights attached to the score is not modelled.
- **Review.** This paper has not been peer reviewed.

10 Conclusion

We reformulated the accountable-witness reputation mechanism of CTP as a consistent model, derived closed-form conditions for its default tolerance, collusion resistance, incentives to contribute and to check, and entry cost, and derived a parameter set from these conditions. Evaluated against baselines, with adaptive agents and under a sensitivity analysis, the recalibrated mechanism makes fraud unprofitable where the original rules collapse. Two failure modes revealed during evaluation were fixed; one applicability condition — witness accuracy — remains. For tontines, the proposed instantiation uses no witnesses but relies on verifiable payments, score-based rotation and sponsorship. The next step is empirical: the data collection described in Appendix B.

Reproducibility. The script `simulations-v2.0/ctp_core_v2_experiments.py` produces all figures, tables and the file `resultats_v2.json`, with fixed seeds.

A Erratum for versions v0.1 and v1.1

1. **Anti-collusion condition (Prop. 4.2).** v1.1 claims that $\phi \geq 3\eta$ suffices for $\tau_w \geq 0.3$ and $\mathbb{P}(\text{detection}) \geq 0.15$. The stated inequality, $\phi > \eta q / (\tau_w \mathbb{P})$, then requires $\phi > 22.2 \eta q$; with $\phi = 3\eta$ it only holds for $q < 0.135$. Moreover, the gain attributed to the colluding witness did not exist in the model.
2. **No incentive to validate.** Fixed by R_i .
3. **Audit targeted at $q(s) < q_{\min}$.** Well-rated coalitions were never audited. Fixed by uniform auditing.
4. **No author sanction.** Fixed by S_i .
5. **Witness sanction divided by $|\mathcal{W}(s)|$.** Diluted accountability. Fixed.
6. **Constant consensus factor.** $\min(|\mathcal{W}(s)|/k_j, 1) = 1$ for every accepted signal. Replaced by $|\mathcal{A}(s)|/k$ with $a_k < k$.
7. **Convergence (Prop. 3.2).** Undefined ϵ ; replaced by Proposition 2.
8. **Inactivity.** Submitting a signal was enough to stay active, which removes the incentive at maximal score (Prop. 7, E6).
9. **Difficulty index ρ_j .** Division by $\tau_{\min}$, which is zero as soon as a member reaches 0; the symbol ρ was used for two objects. Reformulated in Appendix C.
10. **Simulation-based validation.** With $n = 12$, $k = 3$, unanimity and a single accomplice, the success probability of collusion was zero; the code’s audit used the true quality, unknown to the protocol; the honest score of 1.000 reflects saturation.

11. **Sybil resistance.** Claimed without proof. Replaced by Propositions 9 and 11.
12. **Gas costs.** Unmeasured estimates; removed.
13. **References.** Citations in the published v0.1 appeared as “[?]”.

B Data collection protocol

Empirical calibration requires, for each tontine followed and each member-period: the contribution due, the date and amount paid, the pot position, departures and exclusions, as well as group characteristics (size, amount, periodicity, age, existence of guarantors).

Sample size. Estimating a default rate of about 5 % within ± 2 points with 95 % confidence requires about $1.96^2 \times 0.05 \times 0.95 / 0.02^2 \approx 456$ independent member-period observations. Since observations of the same member are correlated, a design effect of 2 raises this to about 900, e.g. three tontines of fifteen members followed over twenty periods.

Target parameters. The observed default rate p_d and the effective tolerance of groups (number of defaults before exclusion) set p_d^* , hence κ/η via Corollary 1; the frequency of departures after receiving the pot allows testing Proposition 10.

Ethics. Informed consent of members, anonymisation of identities and amounts before analysis, and reporting results back to the groups.

C Out-of-scope extensions

v1.1 defined a difficulty index $\rho_j = k_j \bar{\tau}_j / (k_{\min} \tau_{\min})$, undefined when $\tau_{\min} = 0$. A well-defined form is $\rho_j = (k_j / k_{\min}) \cdot (\varepsilon + \bar{\tau}_j) / (\varepsilon + \bar{\tau}_{\mathcal{M}})$, where $\bar{\tau}_{\mathcal{M}}$ is the mean score across all Nodes and $\varepsilon > 0$. This index, the global score and inter-Node portability are not analysed here and should not be considered validated.

D Notation

$\tau_i(t)$	score of member i
k, a_k, θ	panel size, required approvals, required fraction
$\mathcal{P}(s), \mathcal{A}(s)$	panel and approvers of signal s
$q(s), Q(s)$	measured quality, true conformity
η, η_W	author gain, witness reward
κ, κ_s	default penalty, author sanction
δ, ϕ	depreciation, witness accountability
p_a, ε_a, h	audit probability and error, decoy rate
τ_W, τ_0, τ_p	eligibility threshold, founders' and sponsored score
c_h, c_v	cost of an honest contribution, of a check
p_d^*, p_a^*	default-tolerance and audit thresholds
π	panel capture probability
ψ, P	sponsor penalty, probation length

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